Week 01 — Introduction to Internet-of-Things (IoT)

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Contents: 1. What is IoT? · 2. Understanding IoT systems · 3. Practical

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Part 1 — What is IoT?

Learning objectives: explain the meaning of IoT, identify the main components of an IoT system, describe the basic IoT data flow, and recognize IoT applications in daily life and industry.

Opening discussion

> Which of these can be considered IoT? Smart watch · Traffic camera · Factory robot · Delivery tracking system · Air conditioner controlled by smartphone

The answer comes back in Part 3 (Activity 1): **potentially all of them**, if they are **connected and exchange data with other systems**.

Definition and the important idea

> **Internet-of-Things (IoT)** connects the **physical world** with the **digital world**.

- IoT is **not only about connecting devices**.
- **IoT is about using data from physical things**, to answer:
  - What is happening now?
  - Is the condition **normal or abnormal**?
  - Should the system **send an alert**?
  - Should an action be **taken automatically**?

Examples

In daily life	In industry
Smart watch measuring heart rate	**Smart logistics** — status of picking vehicles, empty lots, picking location
Smart home — light and temperature control	**Automated inspection** — scan, inspect, feed back results
Smart parking — parking lot availability	**Energy monitoring** — status and remote control
Smartphone-connected air purifier	**Smart agriculture** — control farm temperature and humidity
Package tracking — location and arrival time	

Main components of IoT

> **IoT = a connected system, not only a single device.**

Component	Role
Physical object	The thing being observed or controlled
Sensor	Collect data from physical objects
Network	Transmit data from sensors to the processing system
Processing system	Receive, analyze, support decisions
Application	Make information available to the user; enable control by users
User or actuator	Acts on the information, or acts physically

Sensors

> **Sensor** = measures and obtains **physical data** from a device.

Sensor type	Data
Temperature sensor	Heat
Humidity sensor	Moisture
Light sensor	Brightness
Camera	Image / video
Accelerometer	Linear motion
Gyroscope	Rotation and angular velocity
GPS	Location

IoT devices

An IoT device usually contains: **sensor, processor, memory, communication module, power source, software**.

Example devices:

- **Smart watch** — sensors, Bluetooth, memory, processing, power source
- **Machine equipped with a sensor**
- **Vehicle tracking device** — GPS, processing, communication, power management

Networks in IoT

IoT devices need communication to **send data and receive commands**. Examples: **Wi-Fi, Bluetooth, Ethernet**.

Network selection depends on: distance, speed, power consumption, cost, reliability.

Processing systems: device, edge, cloud

	Device	Edge	Cloud
Role	Collects raw data	Processes data **near devices**	**Large-scale** storage and analytics
Resources	Very limited CPU/memory	Moderate	Effectively unlimited
Typical tasks	Read temperature, check battery, simple filtering	Event detection, aggregation/filtering	Model training, historical analysis, dashboards, centralized management
Benefit	Low latency at source	Reduced bandwidth, faster response	Store history, analyze trends

> **Key idea: device** handles sensing; **edge** handles nearby processing; **cloud** handles large-scale storage and analytics.

This three-way split is the backbone of the whole course — the second half is about **edge computing**.

IoT in a smart home

- **Gateway:** provides the connection between devices and external services.
- Sensor data is transmitted to a **cloud platform**, which stores information, analyzes it, and makes it available to apps.
- Two directions: **data flow** (sensor → cloud) and **control flow** (commands → device).

IoT in machine monitoring (industry)

```
Temperature/vibration sensors -> Edge gateway / PLC -> Factory network -> Cloud platform -> Monitoring dashboard
                                (filtering,                             (storage, analytics,        (maintenance
                                 aggregation,                            machine learning,           engineer)
                                 protocol connection)                    rule engine)
```

Possible outcomes: real-time monitoring (machine health, performance), **predictive maintenance** (detect anomalies, predict failure), and **alerts** (notify abnormal conditions).

> **Security applies across all layers:** devices, network, cloud, applications.

Why IoT matters

It enables **real-time monitoring, automation, data-driven decision-making, remote control, predictive maintenance, and smart services**.

Part 1 summary

- IoT connects physical objects to digital systems.
- IoT systems **collect, transmit, process and use** data.
- Main components: sensors, devices, networks, processing systems, applications.
- **A device becomes part of IoT when it can:** sense or generate data, communicate through a network, and interact with another digital system or application.
- Applications: homes, cities, factories, healthcare, logistics, transportation.

Part 2 — Understanding IoT systems

Learning objectives: how IoT systems differ from traditional computing systems, their major characteristics and design challenges, and why **scalability, interoperability, reliability and security** matter.

A smart device alone is not a complete IoT system

> Smart watch → Smartphone → Internet → Cloud platform → Health application → User

A complete IoT system includes: **physical device, data, communication, processing, application, user or action**.

The physical–digital loop

Many IoT systems operate as a **continuous loop**:

```
Physical environment -> Sense -> Data -> Process -> Decision -> Action -> Physical environment
```

Example: temperature rises → sensor detects 30 °C → system decides the room is too hot → air conditioner turns on → temperature decreases.

> IoT systems can both **observe** and **control** the physical world.

Traditional system vs IoT system

Traditional system	IoT system
Human enters data	**Sensors generate data**
Mostly digital environment	**Physical + digital**
Usually a few computing devices	**Potentially many devices**
User initiates many actions	**Systems may act automatically**
Computing often centralized	**Computing may be distributed**

Key differences: physical-world interaction, large number of devices, continuous data, heterogeneous hardware, limited computing/storage/communication resources, distributed computing.

Characteristic 1: Many connected devices

From **5 sensors in a room → 500 in a building → 10,000 in a factory → millions in a smart city**.

More devices mean: more data, more communication, more management, **more failures**.

Scalability = can the system continue to work efficiently as the number of devices or amount of data increases? IoT platforms must scale in **devices, messages, storage, processing, and users**.

Characteristic 2: Continuous operation

- 1 sensor at 1 reading/second = **86,400 readings per day**; 1,000 sensors = **86,400,000 readings per day**.
- **Main issue:** do we really need to **send and store every measurement**?

IoT data also **changes over time** (10:00 → 23.5 °C, 10:01 → 24.1 °C, 10:02 → 24.8 °C, 10:03 → 29.7 °C), which raises: when was it measured, is it changing, is the current value abnormal, and is history important?

Characteristic 3: Heterogeneity

IoT systems contain **different types of devices**, differing in manufacturers, sensors, operating systems, **data formats**, communication technologies and processing capabilities.

Example device	Its nature
Smart watch	Small, battery-powered
Security camera	High data volume
Factory machine	High reliability required
Agricultural sensor	Long-distance communication

Different applications have different priorities — IoT architecture is **application-dependent**:

Application	Important requirement
Smart home	Convenience, cost
Healthcare	**Reliability, privacy**
Smart factory	**Fast response, reliability**
Smart agriculture	**Long range, low power**
Smart city	**Scalability**

Characteristic 4: Limited device resources

Many IoT devices have limited **CPU, memory, storage, battery and communication capacity**.

Smartphone / PC	Small IoT sensor
Powerful CPU	Small processor
GB of memory	**KB/MB** of memory
Recharged frequently	**May operate for years**
Fast network	Low-bandwidth network

Energy consumption is a major design issue. Energy is used by **sensor measurement, processing, communication, and keeping the network connection active**.

> **Important principle: sending data can consume significant energy.** More communication → more energy use → shorter battery life.

This is the reason edge filtering exists: **not sending** is often the cheapest optimization.

Characteristic 5: Distributed systems

Processing does not necessarily happen in one place: **device → nearby processing → remote processing**.

Task	Possible location
Read temperature	**Device**
Detect overheating	**Device / Edge**
Store one year of history	**Cloud**
Analyze factory-wide trends	**Cloud**
Trigger immediate alarm	**Device / Edge**

> **Processing location depends on system requirements.** Anything urgent stays local; anything historical goes to the cloud.

The six challenges

#	Challenge	The problem	Responses
1	**Latency**	Delay between an event and the system response. Machine overheats → data to cloud → cloud processes → command returns. **What if this takes several seconds?**	For **safety-critical** applications, **faster local processing** may be required
2	**Bandwidth**	1,000 sensors measuring constantly creates high network traffic, higher cost, more storage and processing	**Raw data → local filtering → send only important data to the cloud**
3	**Network dependency**	If the Internet fails: data cannot be uploaded, remote monitoring stops, decisions may be delayed	**Local processing, local storage, retry mechanisms**
4	**Reliability**	Components fail	Sensor failure → health check / backup sensor; communication failure → retry / local buffering; gateway failure → backup gateway / local operation; low battery → monitoring / power saving; software errors → restart / error handling / update; cloud failure → local processing / later synchronization
5	**Security**	Many possible attack points: unauthorized access, **fake sensor data**, data theft, device takeover, privacy leakage	Security across all layers
6	**Privacy**	IoT may collect location, health information, home activities, video, industrial operation data	**Does all collected data need to leave the local environment? → edge computing**

> **Should one failed component stop the entire system?** No — that is what reliability design is for.

Part 2 summary

IoT systems are challenging because they involve **many devices, continuous data, heterogeneous technologies, limited resources, distributed processing, network dependency, reliability, and security and privacy**.

IoT architecture must consider: where to process, what to send, what to store, how quickly we must respond, and how to keep the system reliable and secure.

Part 3 — Practical

Objectives: decide whether a system is IoT, trace how data moves, distinguish sensing / communication / processing / action, and design a simple conceptual IoT system.

Activity 1: Is it IoT?

Suggested answer: potentially all of them, if they are connected and exchange data with other systems.

Example	Why it may be IoT
Smart watch	Sends sensor data to smartphone/cloud
Smart TV	Communicates with internet services
Traffic camera	Sends traffic data to a monitoring system
Factory robot	Exchanges operational data
Delivery tracker	Sends location/status information
Smart air conditioner	Receives commands from a smartphone

Activity 2: What does IoT data look like?

```
Device ID:   sensor_01
Time:        10:32:15
Temperature: 25.4°C
```

Five things to identify: device identity, measurement time, measured quantity, measured value, **unit**.

Why these fields are needed:

- **Device ID** — which device produced the data?
- **Timestamp** — when was the measurement collected?
- **Sensor value (+ unit)** — what was measured?

This is exactly the record structure used in the week 2 and week 3 practicals.

IoT data changes over time:

```
10:00 sensor_01 23.5°C
10:01 sensor_01 24.1°C
10:02 sensor_01 24.8°C
10:03 sensor_01 29.7°C
10:04 sensor_01 25.2°C
```

- **Which reading is highest?** 10:03, at **29.7 °C**.
- **Why is the timestamp important?** A value alone can't tell you when a condition occurred, how fast it changed, or whether it's current.

- **Why keep previous readings?** To see **trends**, judge whether a value is **abnormal** compared with normal behaviour, and smooth noise (the moving average in week 2).

Activity 3: Trace an IoT system (smart home)

> A temperature sensor measures room temperature. Data is sent via **Wi-Fi** to a smart home application. The user views the temperature and controls the air conditioner.

Data flow	Action/control flow
Temperature → Sensor → Network → Application → User	User → Application → Network → Air conditioner

> IoT systems can both **observe** and **control** the physical world.

Activity 4: Arrange the components (smart health monitoring)

Suggested answer:

```
Patient -> Wearable sensor -> Smartphone -> Internet -> Cloud platform -> Doctor dashboard
```

- **Where is the physical world?** The patient.
- **Where does sensing occur?** The wearable sensor.
- **Where is communication used?** Sensor → smartphone, and smartphone → internet → cloud.
- **Who uses the information?** The doctor (through the dashboard), and the patient through their app.

Activity 5: Smart home, healthcare and industry

	Smart home	Healthcare	Industry
Physical object	Room / home	Patient	Machine
Sensor	Temperature sensor	Heart-rate sensor	Vibration sensor
Data	Temperature	Heart rate	Vibration
User	Resident	Doctor	Engineer
Possible action	Control AC	Send alert	Maintenance

> **The applications are different, but the basic structure is similar.**

Activity 6: Design your own IoT system

Example — smart classroom. *Problem:* the classroom becomes too hot.

```
Room temperature -> Temperature sensor -> IoT device -> Wi-Fi -> Monitoring system
                 -> Teacher / building manager -> Air conditioner
```

```
Device: classroom_sensor_01
Time:   14:20
Temperature: 29.5°C
```

Display in the application: **Classroom A — Temperature: 29.5 °C**

Group task (3–5 members): choose smart home, smart farm, smart parking, smart factory or smart logistics, then answer: (1) what problem are you solving, (2) what should be measured, (3) build the system.

Practice roadmap

Week	Practical topic
1	Understand IoT systems without programming
2	Generate and process sensor measurements
3	Represent IoT data
4–6	Connect and communicate
7–15	Build increasingly capable IoT and edge systems

Preparation for week 2: form groups of 3–4, bring a laptop, install **Python 3.+**, prepare an editor, and check that `print("Hello IoT")` runs.

Most important for exams

- **IoT connects the physical and digital worlds, and is about *using data from physical things*** — not merely connecting devices. The four questions: what's happening now, normal or abnormal, alert, automatic action.
- **A device is part of IoT when it can: sense/generate data, communicate over a network, and interact with another digital system.**
- **Components in order:** physical object → **sensor** → **network** → **processing system** → **application** → **user/actuator**.
- **Sensor table:** temperature/humidity/light/camera/accelerometer/gyroscope/GPS and what each measures.
- **Device vs edge vs cloud:** sensing / nearby processing / large-scale storage and analytics. Be ready to place tasks (read temperature = device, immediate alarm = device-edge, one year of history = cloud, factory-wide trends = cloud).
- **Physical–digital loop:** sense → data → process → decision → action.
- **Traditional vs IoT system table** (human-entered vs sensor-generated data, digital vs physical+digital, few vs many devices, user-initiated vs automatic, centralized vs distributed).
- **Five characteristics:** many devices (**scalability**), continuous operation, **heterogeneity**, limited resources/energy, distributed processing.
- **Six challenges:** latency, bandwidth, network dependency, reliability, security, privacy — with the standard response to each.
- **Sending data costs energy:** more communication → more energy → shorter battery life. Filtering locally saves bandwidth, energy and cost.
- **Application priorities:** home = convenience/cost, healthcare = reliability/privacy, factory = fast response/reliability, agriculture = long range/low power, city = scalability.
- **An IoT data record needs:** device ID, timestamp, quantity, value, unit.